

EX.NO : 08

DATE :

DESIGNING HYBRID TOPOLOGIES USING CISCO PACKET TRACER

AIM :

To Designing a Hybrid topologies by using Cisco Packet Tracer.

REQUIREMENTS :

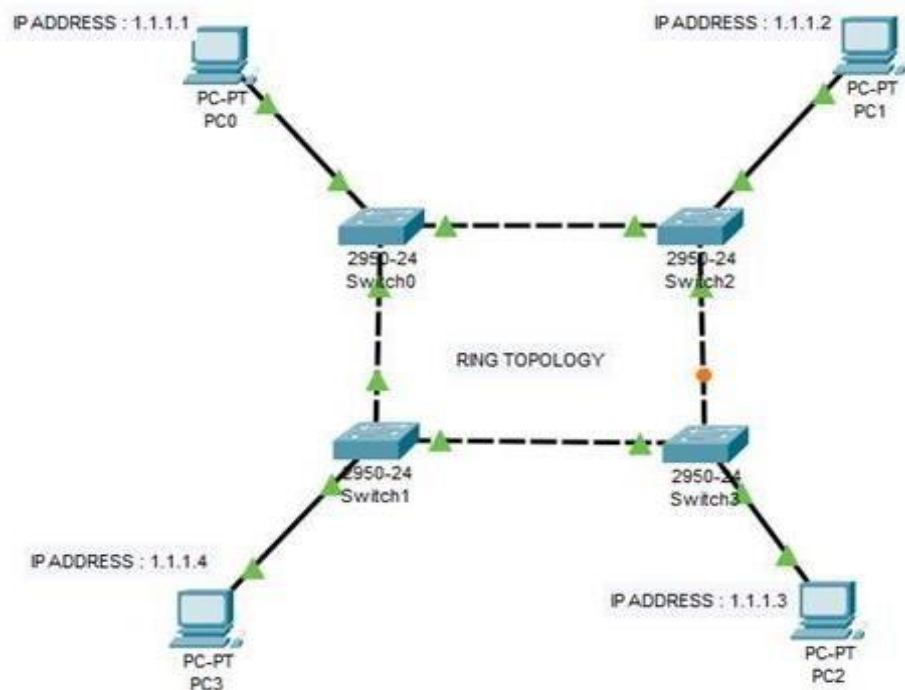
- 16 windows PC.
- 12 Switch (2950-24).
- 16 Straight Line LAN Cables.
- 13 Cross Over Cables.
- Cisco Packet Tracer.

PROCEDURES :

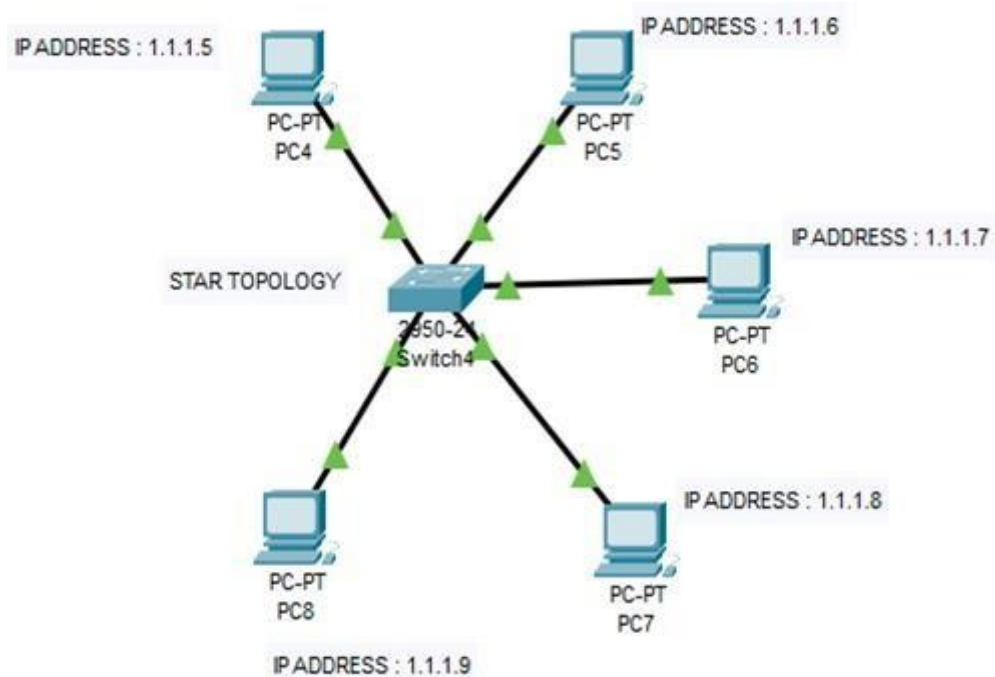
- Open CISCO PACKET TRACER software.
- Draw the 16 PC using END Device Icons.
- Draw the 12 Cisco 2950-24 Switch Using Switch icon lists.
- Make the Connections using Copper-Straight-Through Ethernet Cables.
- Make the Connections between Switches using Cross Overs Cables.
- Enter the IP Address To Each Machine.
- Check the Network Connections using Add Simple PDU(P).

NETWORK TOPOLOGY

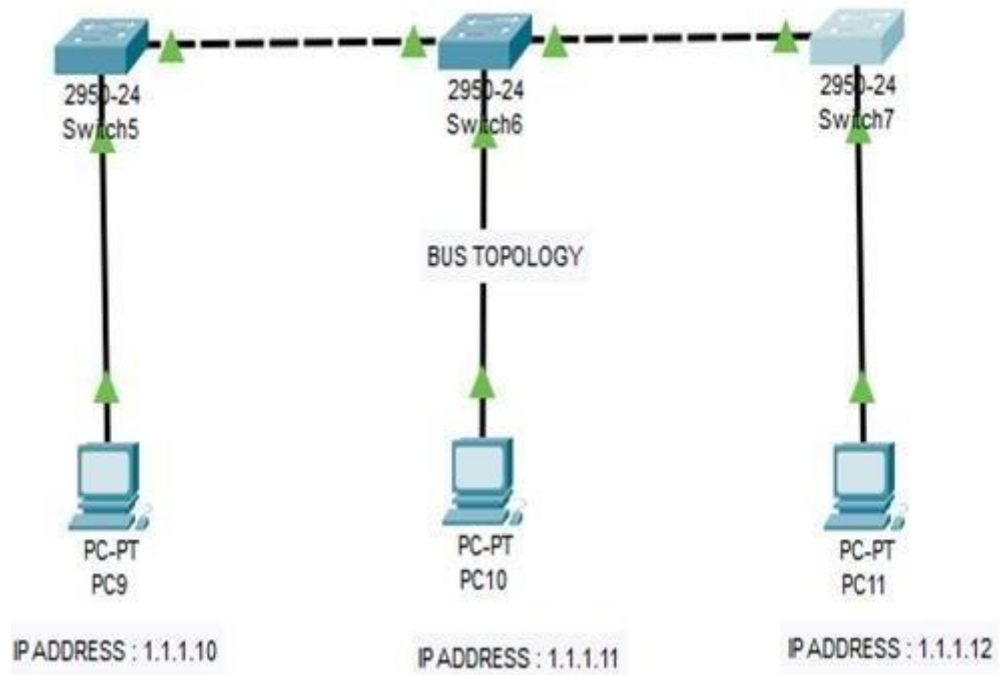
RING TOPOLOGY



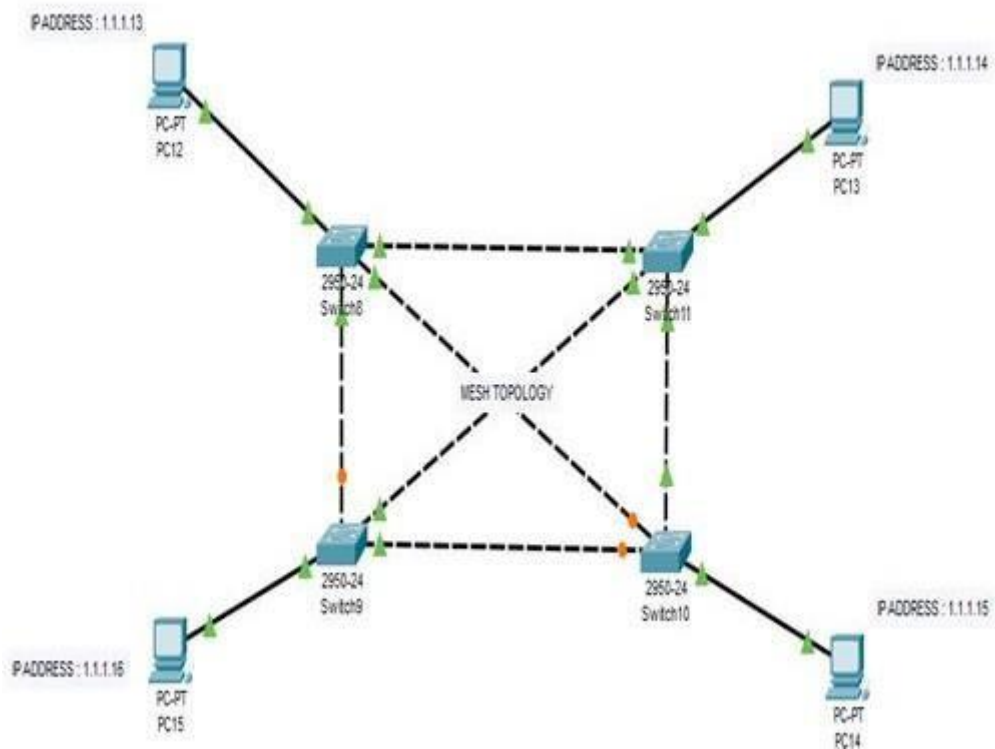
STAR TOPOLOGY



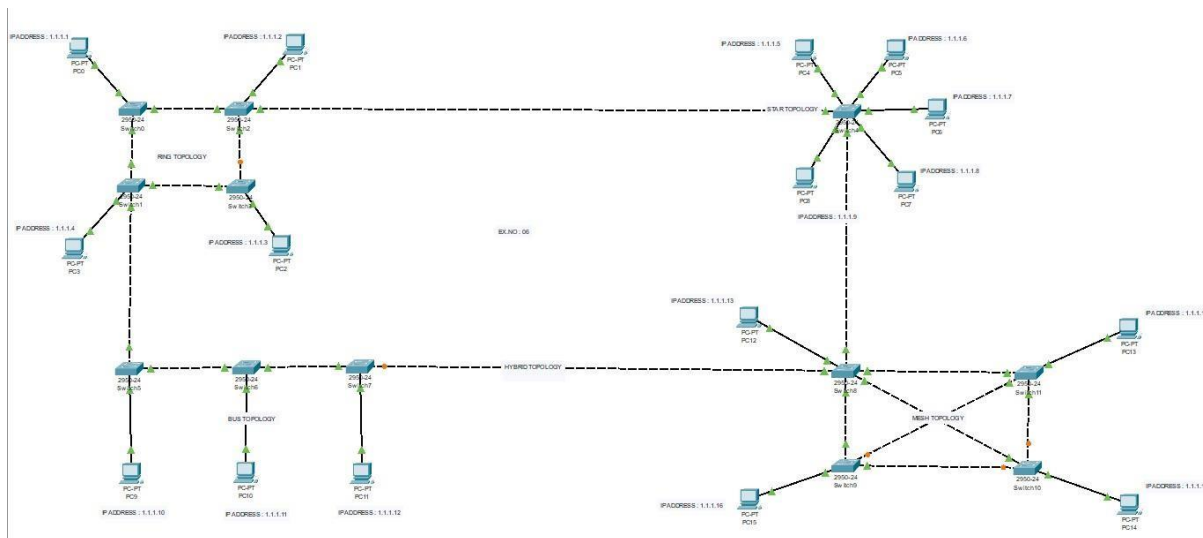
BUS TOPOLOGY



MESH TOPOLOGY



HYBRID TOPOLOGY



HOST PC IP ADDRESS :

PC0 IP CONFIGURATION

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 1.1.1.1

Subnet Mask: 255.0.0.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

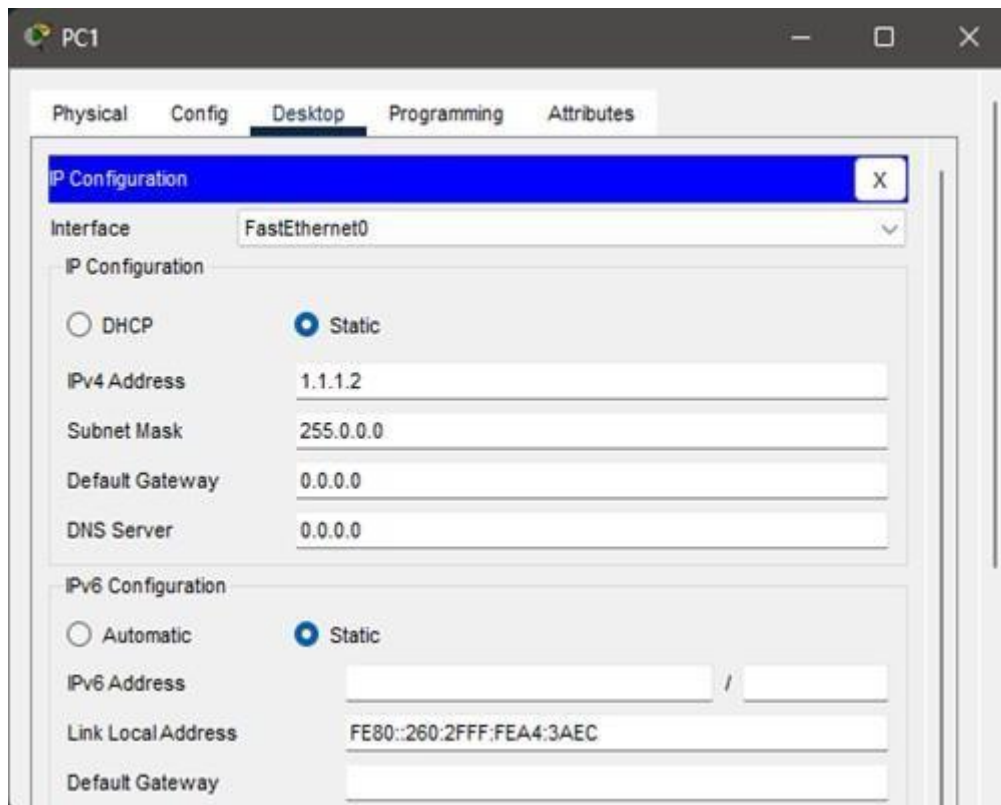
☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::210:11FF:FEC3:70B3

Default Gateway:

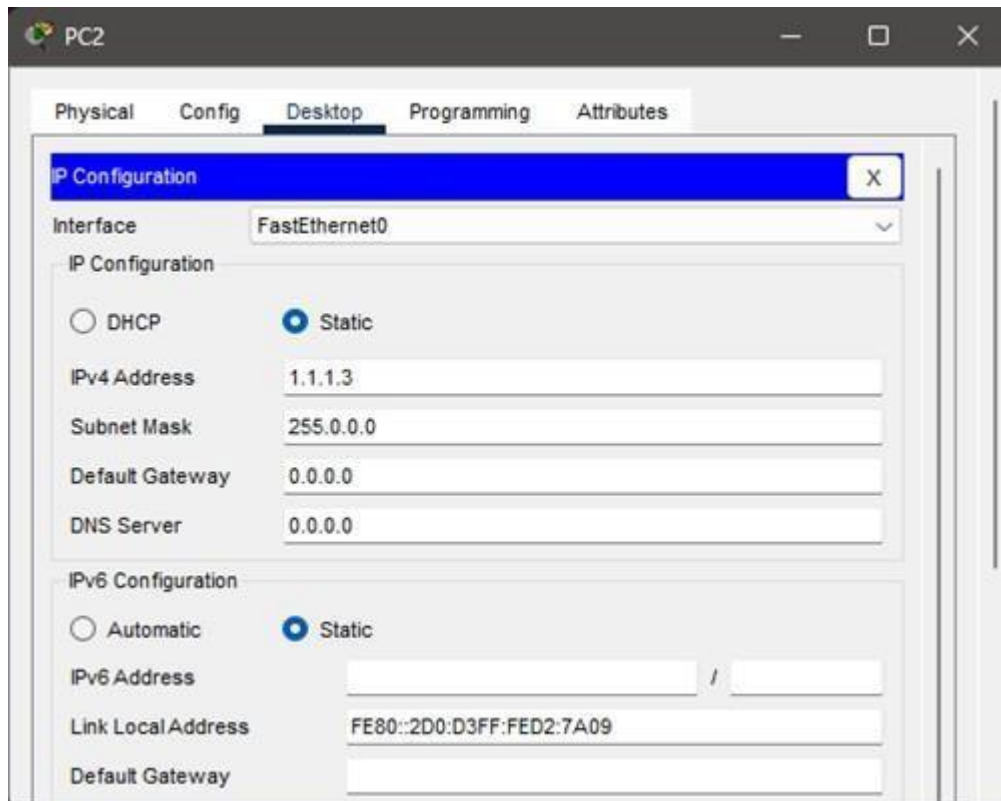
PC1 IP CONFIGURATION



The image shows the 'PC1' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The 'IPv4 Address' is '1.1.1.2', 'Subnet Mask' is '255.0.0.0', 'Default Gateway' is '0.0.0.0', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', 'Static' is selected. The 'IPv6 Address' is empty, 'Link Local Address' is 'FE80::260:2FFF:FEA4:3AEC', and 'Default Gateway' is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.2
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::260:2FFF:FEA4:3AEC
Default Gateway	

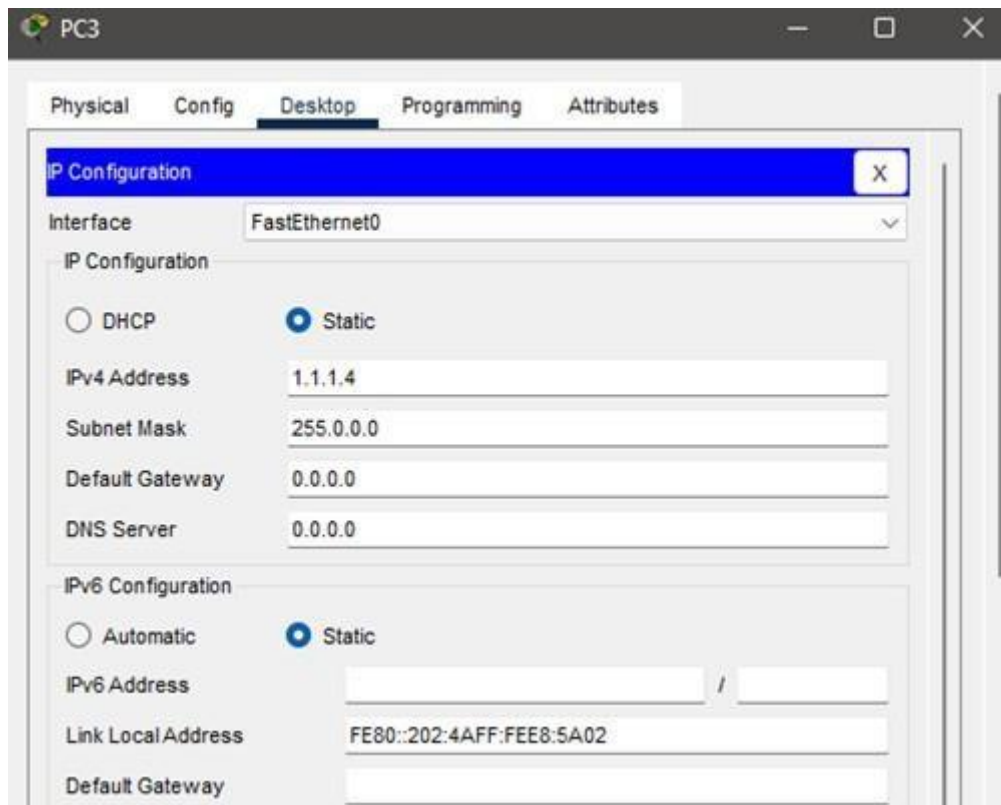
PC2 IP CONFIGURATION



The image shows the 'PC2' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The 'IPv4 Address' is '1.1.1.3', 'Subnet Mask' is '255.0.0.0', 'Default Gateway' is '0.0.0.0', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', 'Static' is selected. The 'IPv6 Address' is empty, 'Link Local Address' is 'FE80::2D0:D3FF:FED2:7A09', and 'Default Gateway' is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.3
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::2D0:D3FF:FED2:7A09
Default Gateway	

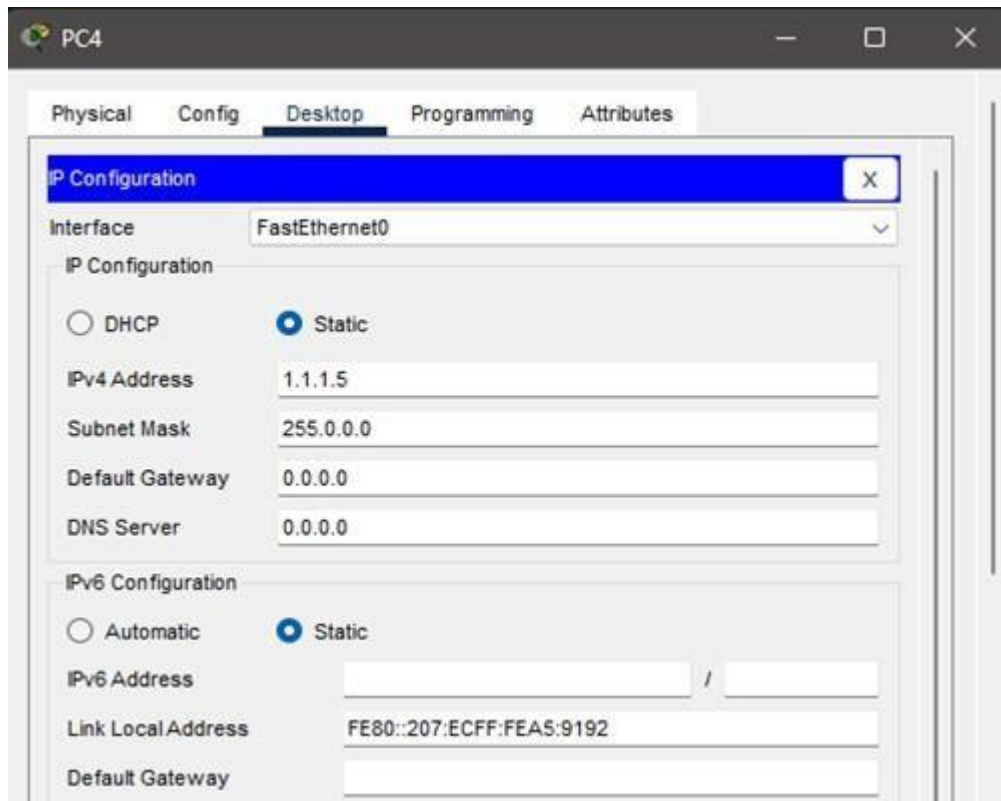
PC3 IP CONFIGURATION



The image shows the 'IP Configuration' window for PC3. The window has a title bar with 'PC3' and standard minimize, maximize, and close buttons. Below the title bar are tabs for 'Physical', 'Config', 'Desktop' (selected), 'Programming', and 'Attributes'. The 'IP Configuration' window is a sub-dialog with a blue header and a close button. It shows the 'Interface' as 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The 'IPv4 Address' is '1.1.1.4', 'Subnet Mask' is '255.0.0.0', 'Default Gateway' is '0.0.0.0', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', the 'Static' radio button is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is 'FE80::202:4AFF:FEE8:5A02', and 'Default Gateway' is empty.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.4
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::202:4AFF:FEE8:5A02
Default Gateway	

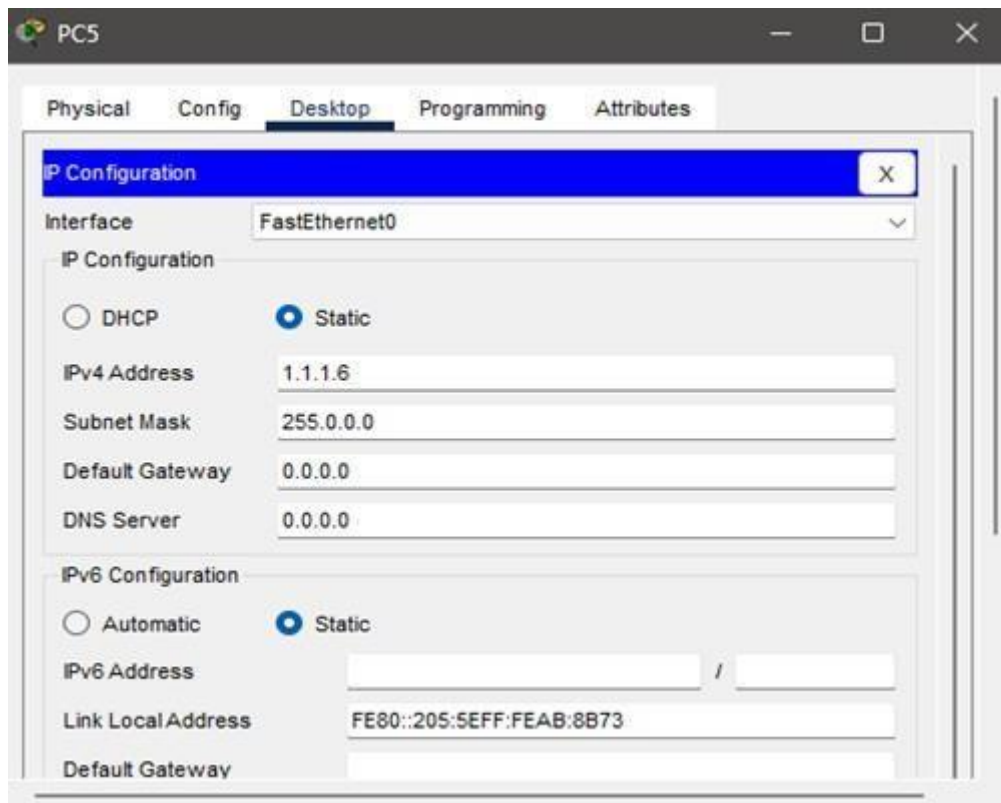
PC4 IP CONFIGURATION



The image shows the 'IP Configuration' window for PC4. The window has a title bar with 'PC4' and standard minimize, maximize, and close buttons. Below the title bar are tabs for 'Physical', 'Config', 'Desktop' (selected), 'Programming', and 'Attributes'. The 'IP Configuration' window is a sub-dialog with a blue header and a close button. It shows the 'Interface' as 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The 'IPv4 Address' is '1.1.1.5', 'Subnet Mask' is '255.0.0.0', 'Default Gateway' is '0.0.0.0', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', the 'Static' radio button is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is 'FE80::207:ECFF:FEA5:9192', and 'Default Gateway' is empty.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.5
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::207:ECFF:FEA5:9192
Default Gateway	

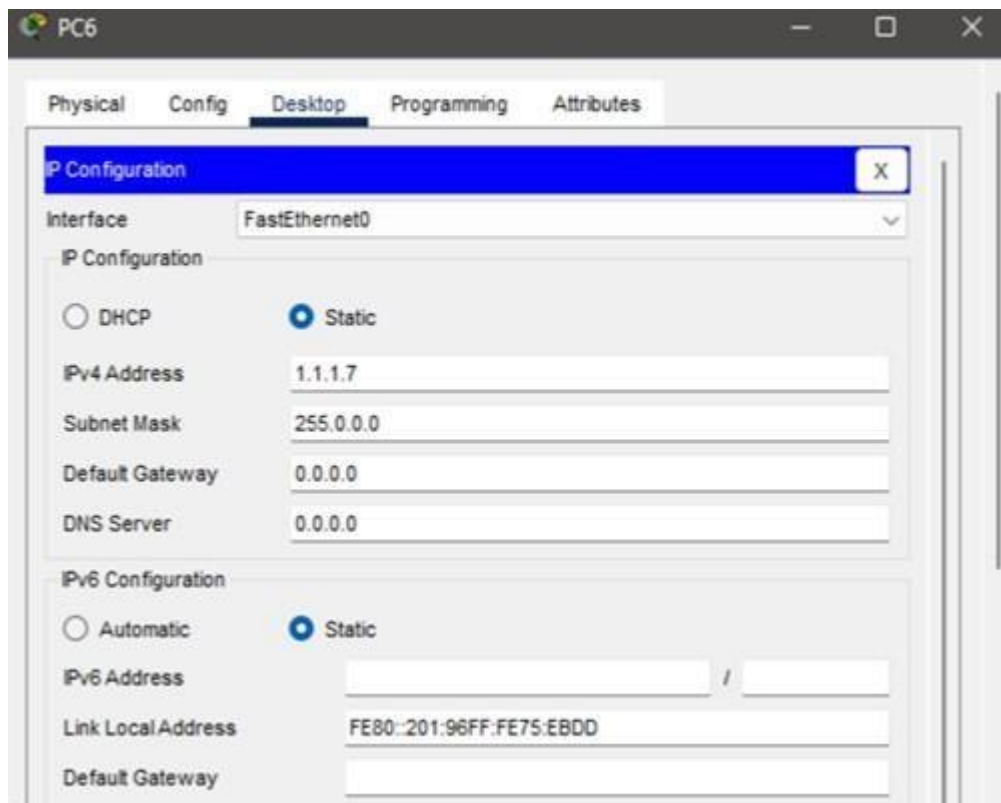
PC5 IP CONFIGURATION



The image shows the 'PC5' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 Address is '1.1.1.6', Subnet Mask is '255.0.0.0', Default Gateway is '0.0.0.0', and DNS Server is '0.0.0.0'. Under 'IPv6 Configuration', the 'Static' radio button is selected. The IPv6 Address field is empty, the Link Local Address is 'FE80::205:5EFF:FEAB:8B73', and the Default Gateway is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.6
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::205:5EFF:FEAB:8B73
Default Gateway	

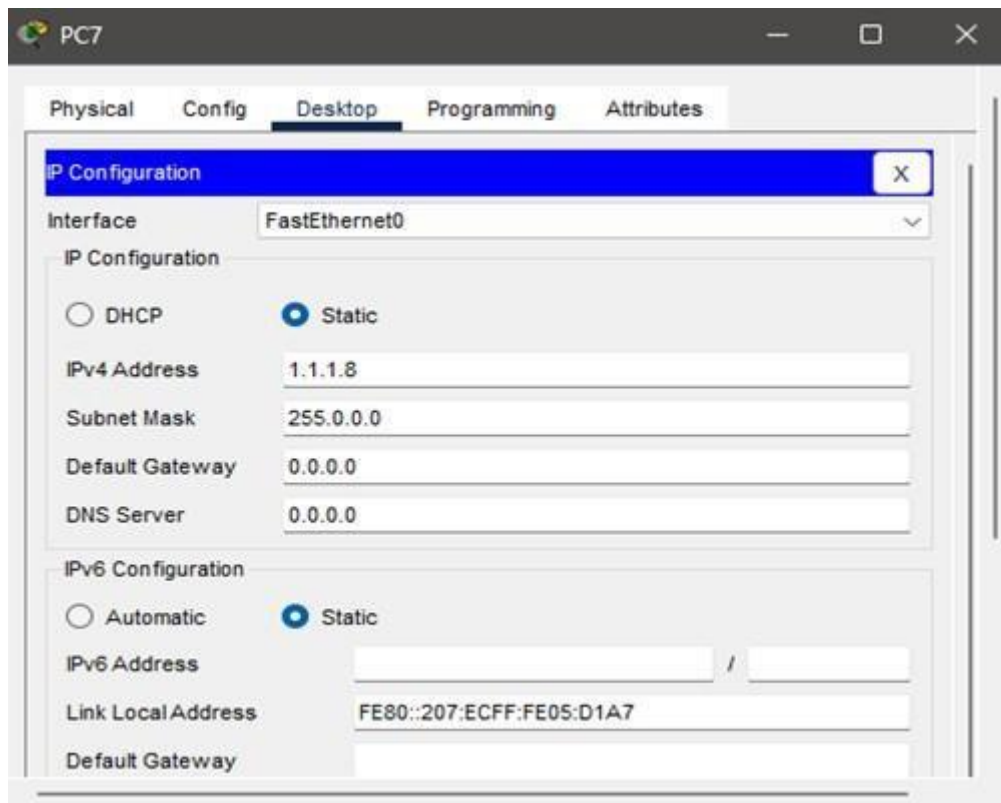
PC6 IP CONFIGURATION



The image shows the 'PC6' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 Address is '1.1.1.7', Subnet Mask is '255.0.0.0', Default Gateway is '0.0.0.0', and DNS Server is '0.0.0.0'. Under 'IPv6 Configuration', the 'Static' radio button is selected. The IPv6 Address field is empty, the Link Local Address is 'FE80::201:96FF:FE75:EBDD', and the Default Gateway is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.7
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::201:96FF:FE75:EBDD
Default Gateway	

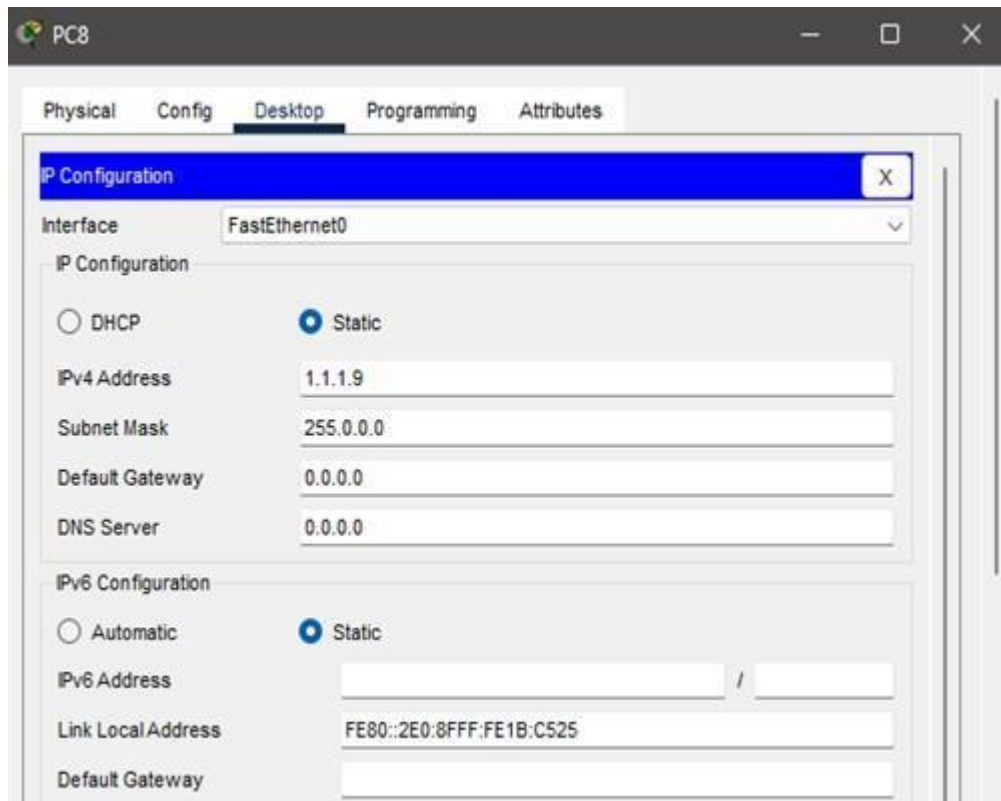
PC7 IP CONFIGURATION



The screenshot shows the 'IP Configuration' window for PC7. The 'Desktop' tab is selected. The 'Interface' is 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IPv4 Address is '1.1.1.8', Subnet Mask is '255.0.0.0', Default Gateway is '0.0.0.0', and DNS Server is '0.0.0.0'. Under 'IPv6 Configuration', 'Static' is selected. The IPv6 Address is empty, Link Local Address is 'FE80::207:ECFF:FE05:D1A7', and Default Gateway is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.8
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::207:ECFF:FE05:D1A7
Default Gateway	

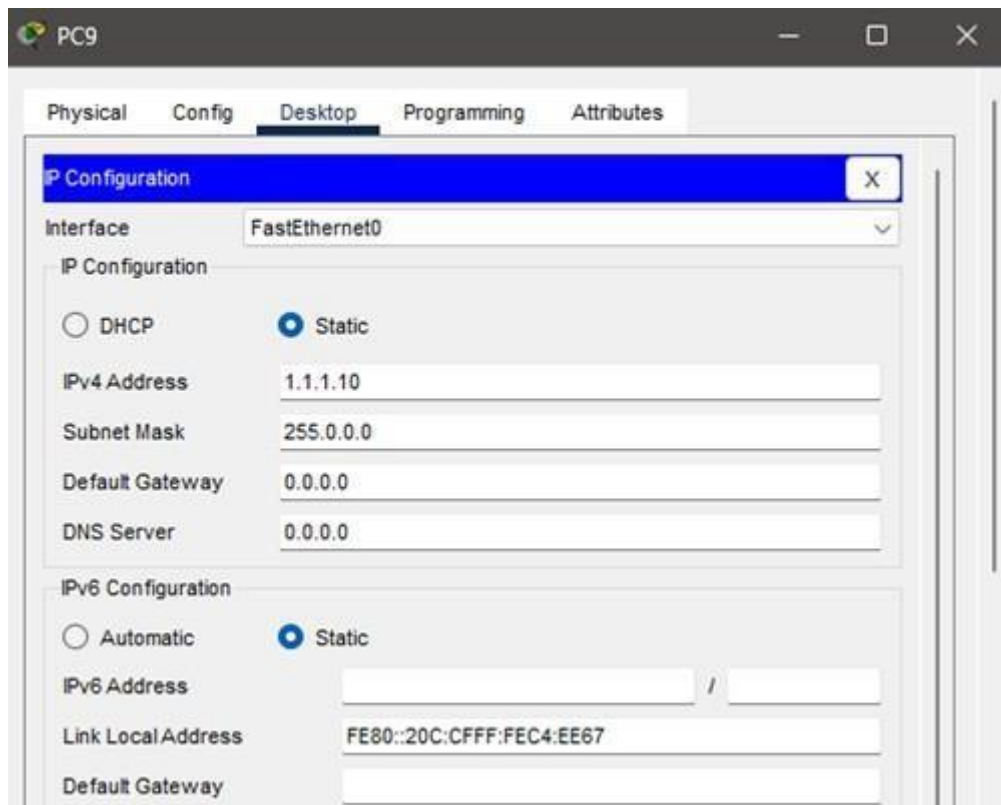
PC8 IP CONFIGURATION



The screenshot shows the 'IP Configuration' window for PC8. The 'Desktop' tab is selected. The 'Interface' is 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IPv4 Address is '1.1.1.9', Subnet Mask is '255.0.0.0', Default Gateway is '0.0.0.0', and DNS Server is '0.0.0.0'. Under 'IPv6 Configuration', 'Static' is selected. The IPv6 Address is empty, Link Local Address is 'FE80::2E0:8FFF:FE1B:C525', and Default Gateway is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.9
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::2E0:8FFF:FE1B:C525
Default Gateway	

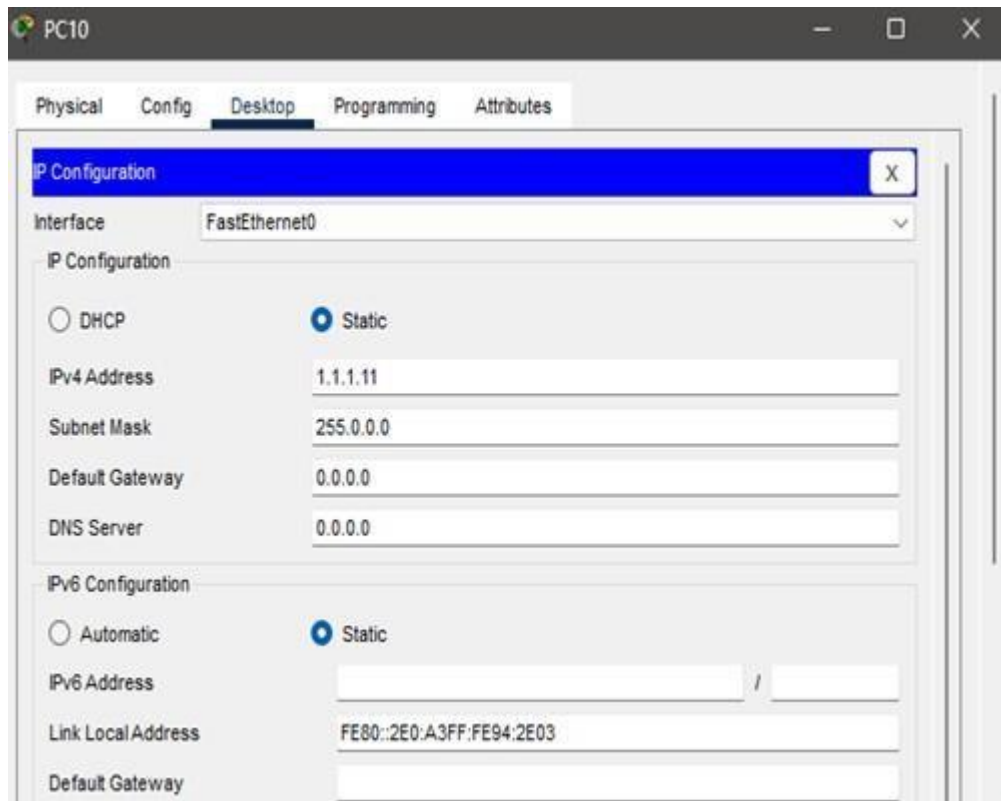
PC9 IP CONFIGURATION



The screenshot shows the 'PC9' window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The 'IPv4 Address' is '1.1.1.10', 'Subnet Mask' is '255.0.0.0', 'Default Gateway' is '0.0.0.0', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', the 'Static' radio button is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is 'FE80::20C:CFFF:FEC4:EE67', and 'Default Gateway' is empty.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.10
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::20C:CFFF:FEC4:EE67
Default Gateway	

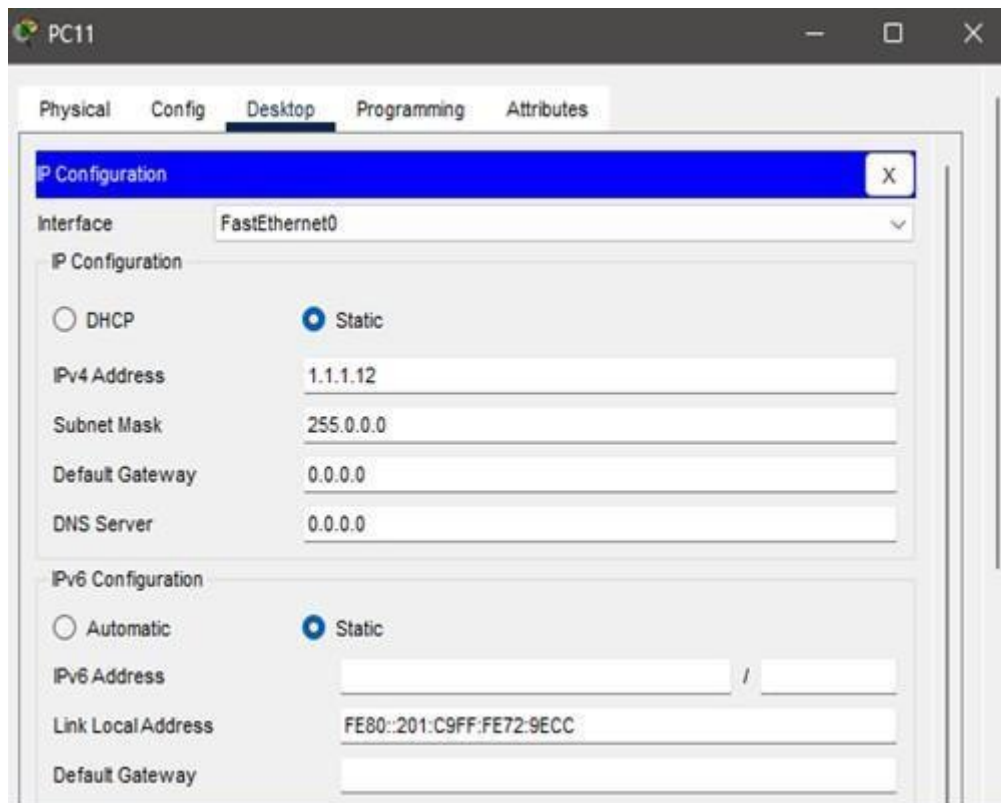
PC10 IP CONFIGURATION



The screenshot shows the 'PC10' window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The 'IPv4 Address' is '1.1.1.11', 'Subnet Mask' is '255.0.0.0', 'Default Gateway' is '0.0.0.0', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', the 'Static' radio button is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is 'FE80::2E0:A3FF:FE94:2E03', and 'Default Gateway' is empty.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.11
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::2E0:A3FF:FE94:2E03
Default Gateway	

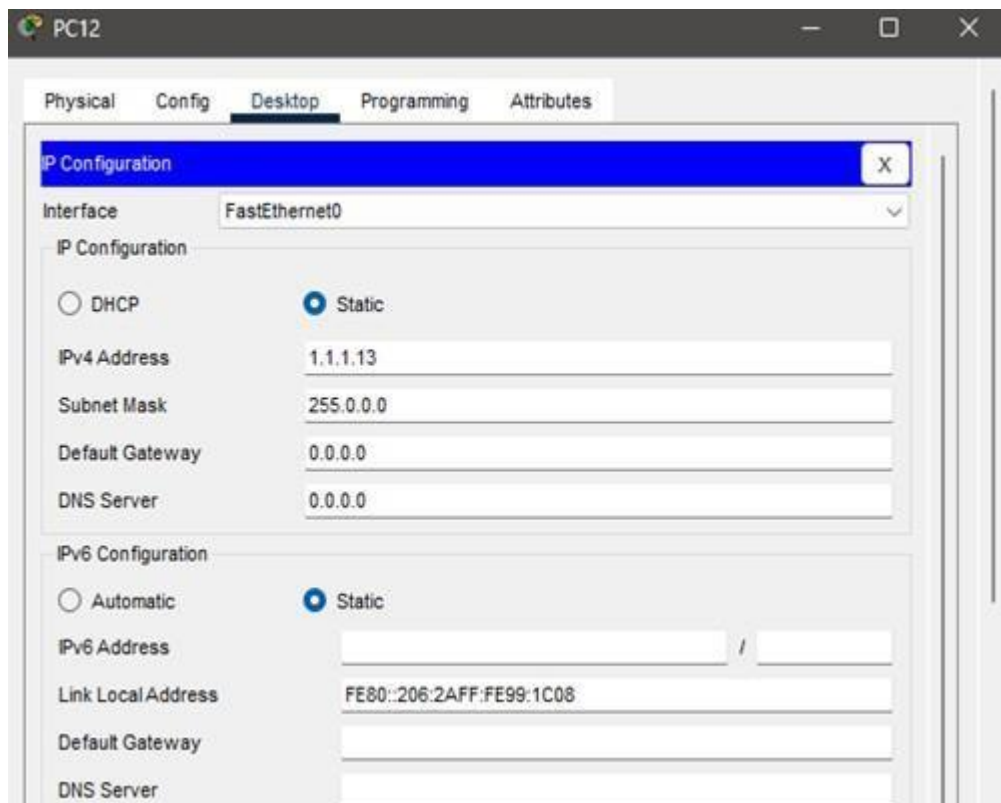
PC11 IP CONFIGURATION



The image shows the 'PC11' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 settings are: IPv4 Address: 1.1.1.12, Subnet Mask: 255.0.0.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The IPv6 settings are: IPv6 Address: (empty), Link Local Address: FE80::201:C9FF:FE72:9ECC, and Default Gateway: (empty).

Section	Field	Value
IP Configuration	Interface	FastEthernet0
	DHCP	☐
	Static	☒
	IPv4 Address	1.1.1.12
	Subnet Mask	255.0.0.0
IPv6 Configuration	Automatic	☐
	Static	☒
	IPv6 Address	
	Link Local Address	FE80::201:C9FF:FE72:9ECC
	Default Gateway	

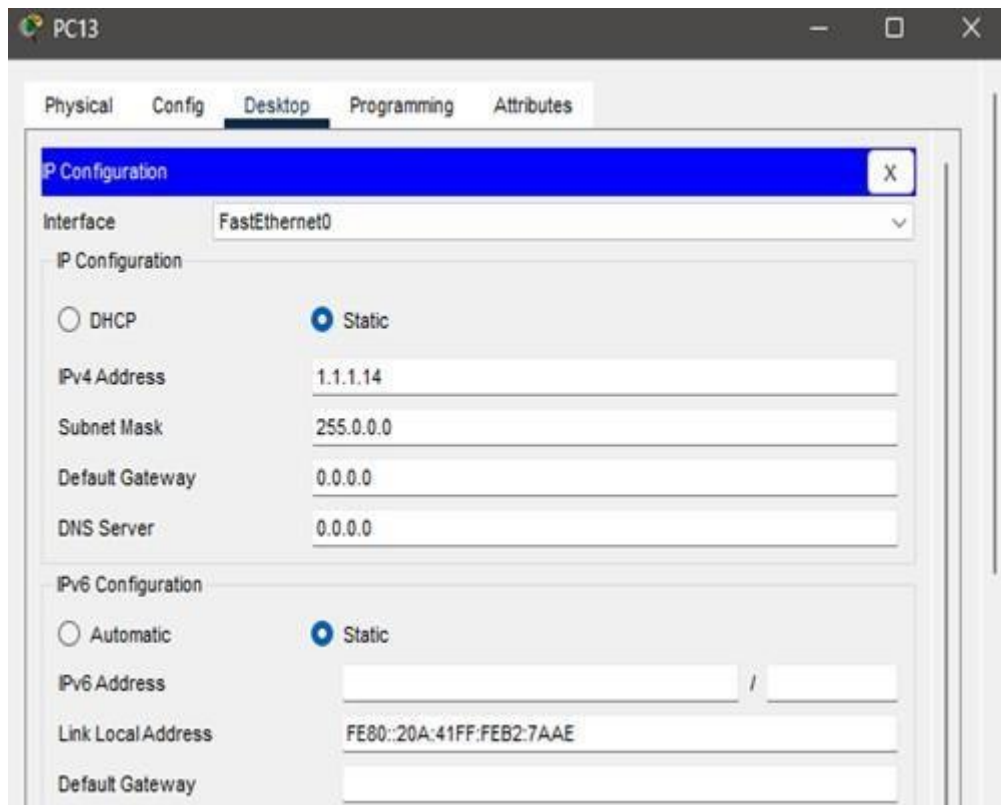
PC12 IP CONFIGURATION



The image shows the 'PC12' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 settings are: IPv4 Address: 1.1.1.13, Subnet Mask: 255.0.0.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The IPv6 settings are: IPv6 Address: (empty), Link Local Address: FE80::206:2AFF:FE99:1C08, and Default Gateway: (empty).

Section	Field	Value
IP Configuration	Interface	FastEthernet0
	DHCP	☐
	Static	☒
	IPv4 Address	1.1.1.13
	Subnet Mask	255.0.0.0
IPv6 Configuration	Automatic	☐
	Static	☒
	IPv6 Address	
	Link Local Address	FE80::206:2AFF:FE99:1C08
	Default Gateway	

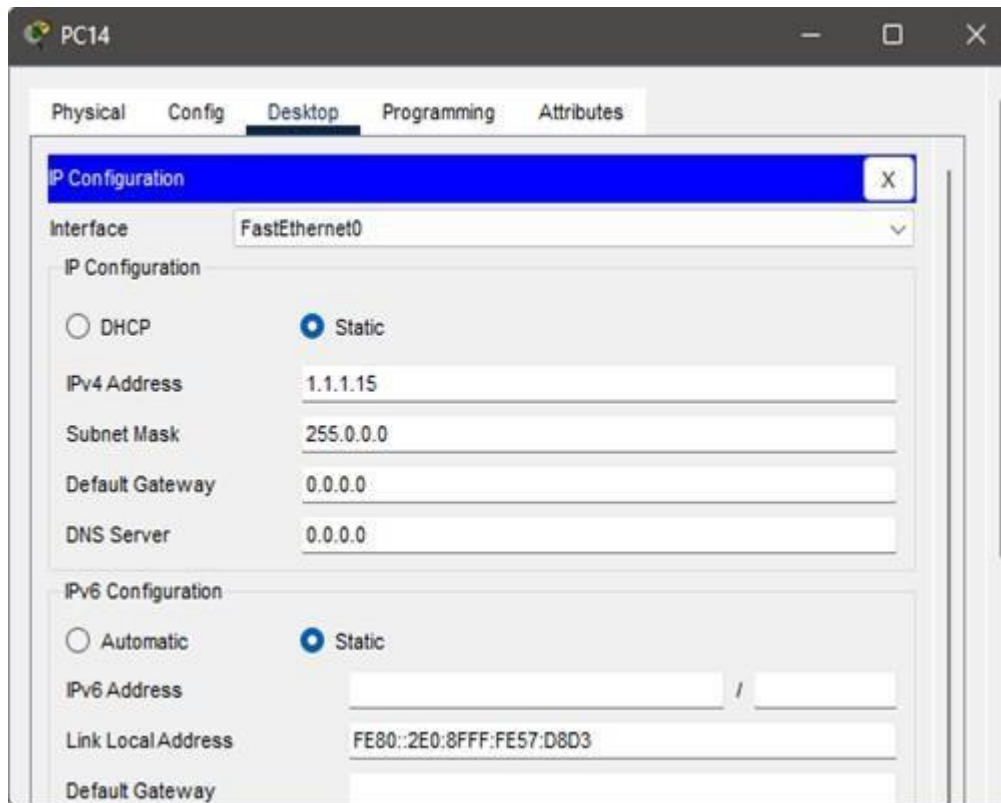
PC13 IP CONFIGURATION



The screenshot shows the 'IP Configuration' window for PC13. The 'Desktop' tab is selected. The 'Interface' is 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IPv4 Address is '1.1.1.14', Subnet Mask is '255.0.0.0', Default Gateway is '0.0.0.0', and DNS Server is '0.0.0.0'. Under 'IPv6 Configuration', 'Static' is selected. The IPv6 Address field is empty, Link Local Address is 'FE80::20A:41FF:FE82:7AAE', and the Default Gateway is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.14
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::20A:41FF:FE82:7AAE
Default Gateway	

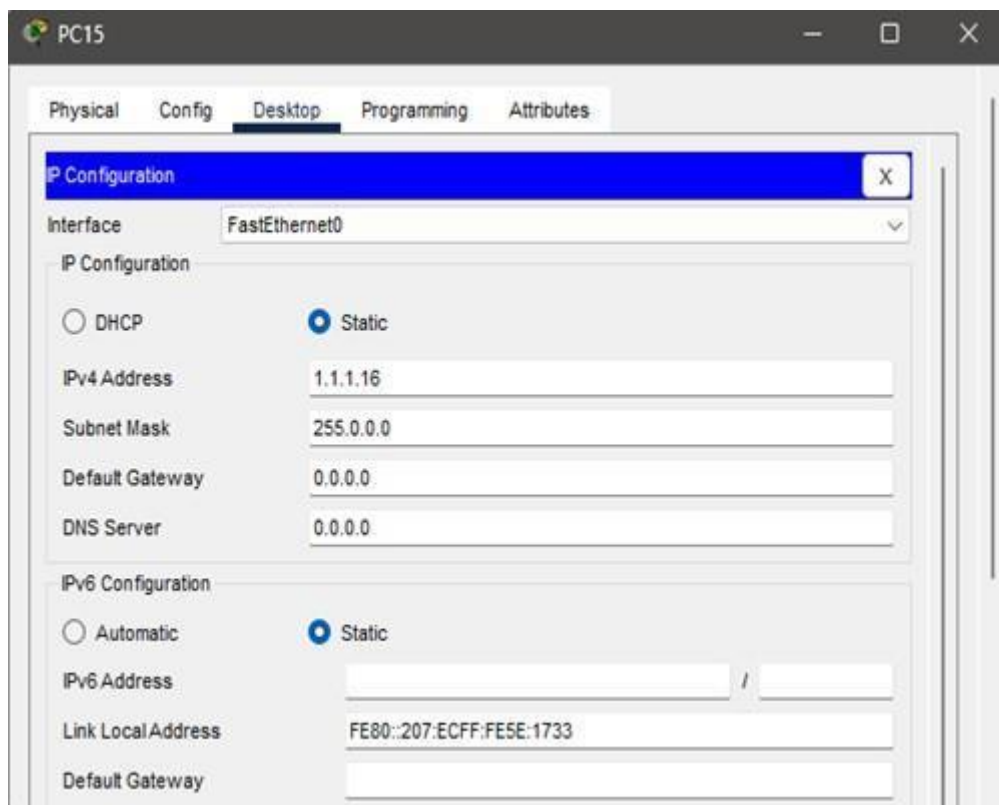
PC14 IP CONFIGURATION



The screenshot shows the 'IP Configuration' window for PC14. The 'Desktop' tab is selected. The 'Interface' is 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The IPv4 Address is '1.1.1.15', Subnet Mask is '255.0.0.0', Default Gateway is '0.0.0.0', and DNS Server is '0.0.0.0'. Under 'IPv6 Configuration', 'Static' is selected. The IPv6 Address field is empty, Link Local Address is 'FE80::2E0:8FFF:FE57:D8D3', and the Default Gateway is empty.

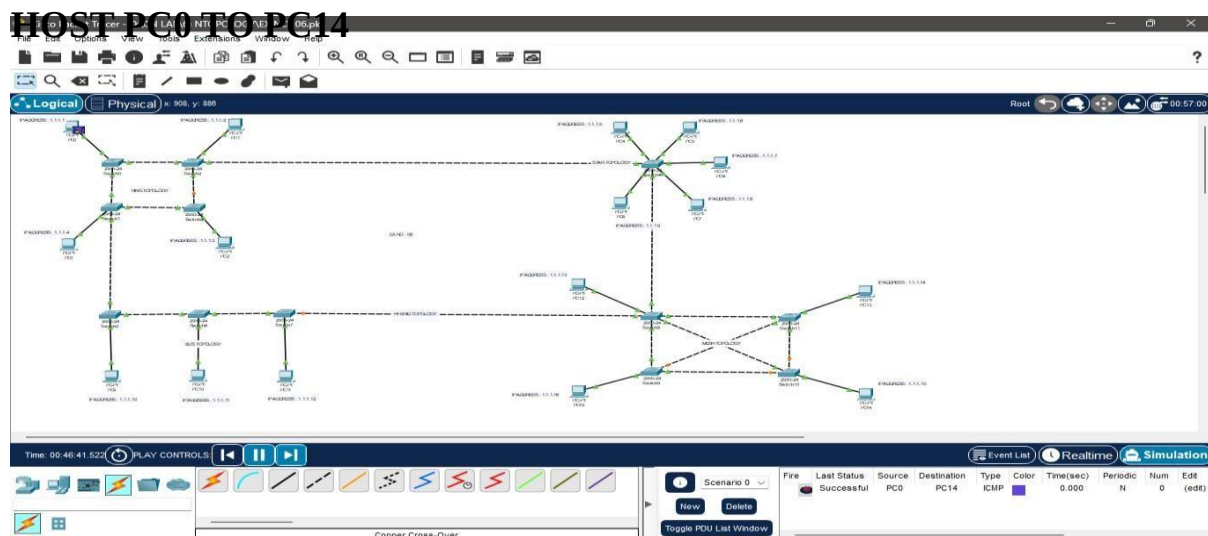
Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	1.1.1.15
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::2E0:8FFF:FE57:D8D3
Default Gateway	

PC15 IP CONFIGURATION

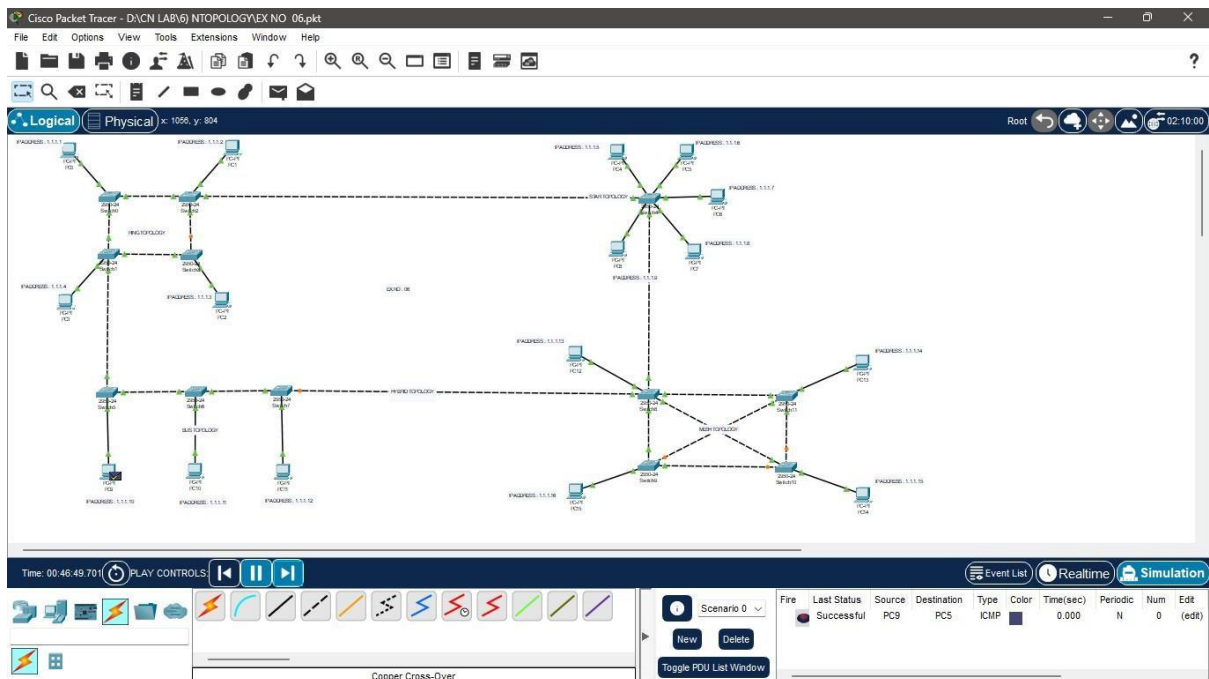


VERIFY LAN NETWORK CONNECTIVITY :

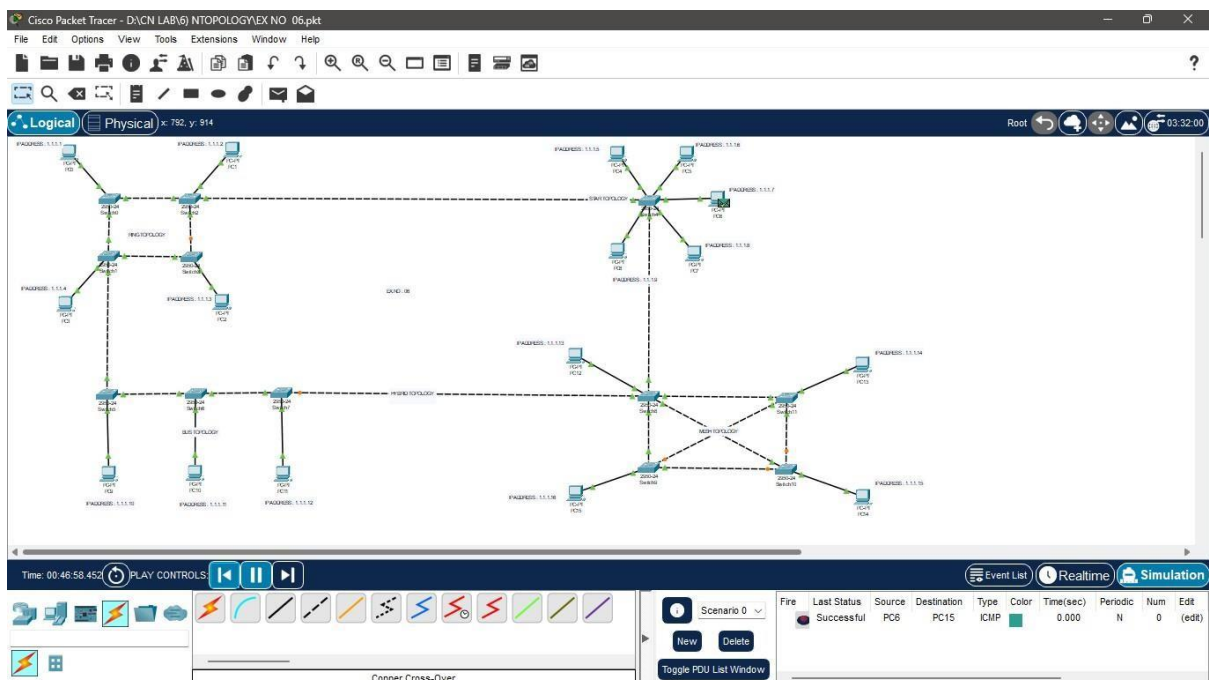
Using Add Simple PDU(p), Click the mail icon and then drop one mail to one of the PC in first lan and another mail to PC in another lan. If the resultant window shows the successful delivery, then network connectivity is successful.



HOST PC9 TO PC5



HOST PC6 TO PC15



RESULT :

Thus, Hybrid topologies are designed using cisco packet tracer and the communication between Hybrid topologies is checked successfully